Name : Mihir Mungara

Airflow : Assignment-1

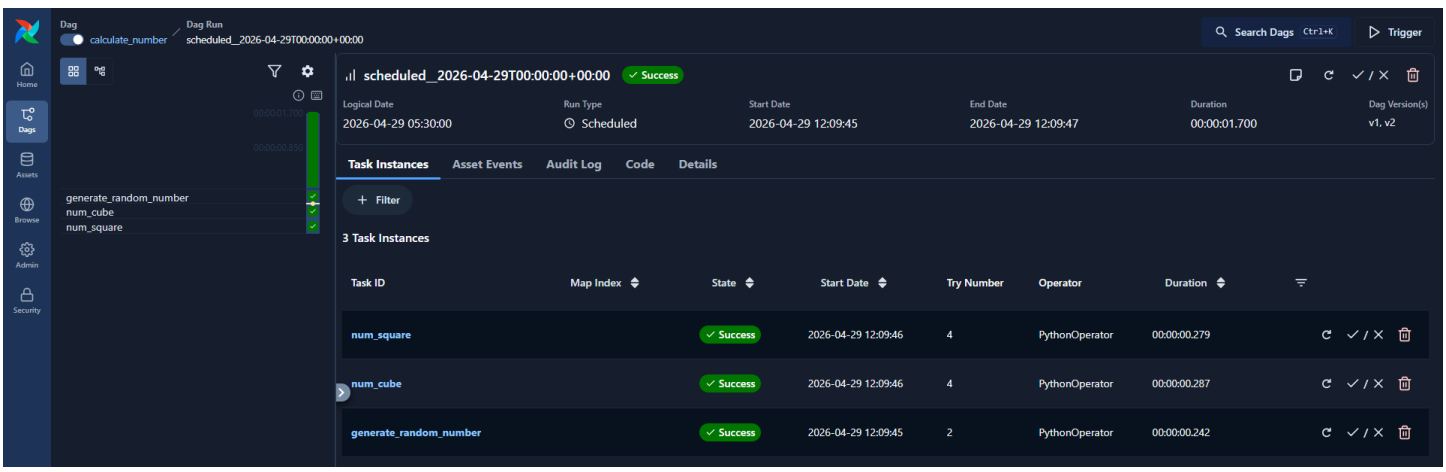
Definition :- Create DAG 'calculate_number' where first task will generate a random number which will be forwarded to 2 downstream task, One will calculate Square and Another will calculate Cube of number, Both tasks will run parallelly as downstream task of generate_number.

DAG Code :-

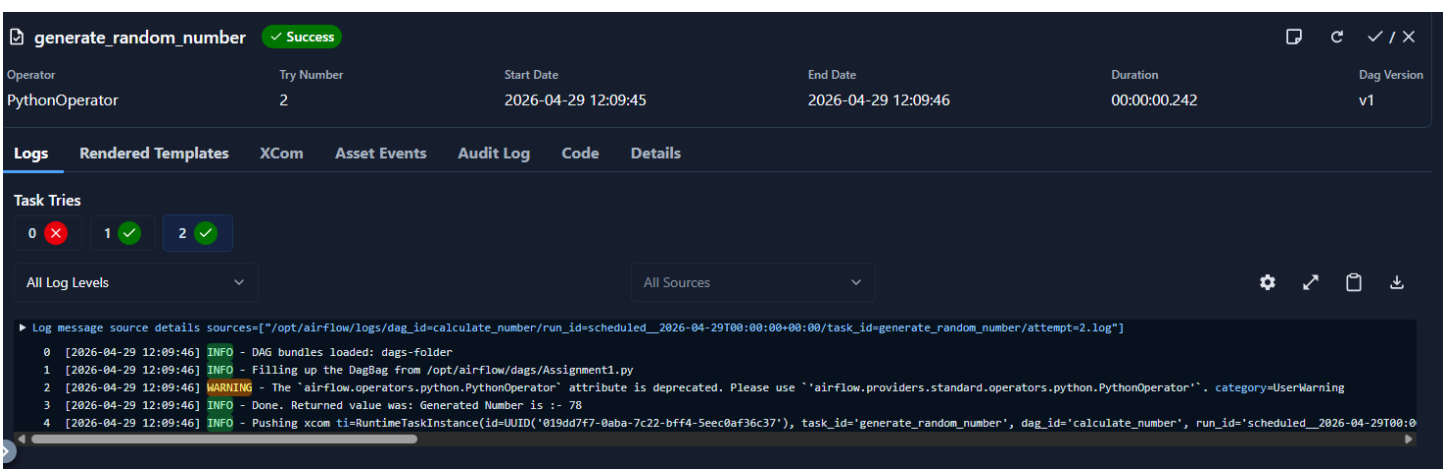
```
from datetime import datetime, timedelta
import random
from airflow import DAG
from airflow.operators.python import PythonOperator

default_args = {
    'owner' : 'mihir',
    'retries' : 5,
    'retry_delay' : timedelta(minutes=2)
}

def generate_random_number(ti):
    num = random.randint(1,100)
    ti.xcom_push(key = 'number', value = num)
    return num

def num_square(ti):
    sq = ti.xcom_pull(task_ids = 'generate_random_number', key='number')
    return sq * sq

def num_cube(ti):
    cb = ti.xcom_pull(task_ids = 'generate_random_number', key='number')
    return cb * cb * cb

with DAG(
    dag_id = 'calculate_number',
    default_args = default_args,
    description = 'this dag is created for calculating number then square and cube of the
same',
    start_date = datetime(2026, 4, 25, 2),
    schedule = '@daily'
) as dag:
    task1 = PythonOperator(
```

```
        task_id = 'generate_random_number',
        python_callable=generate_random_number
    )

    task2 = PythonOperator(
        task_id='num_square',
        python_callable=num_square
    )

    task3 = PythonOperator(
        task_id='num_cube',
        python_callable=num_cube
    )

    task1 >> [task2, task3]
```

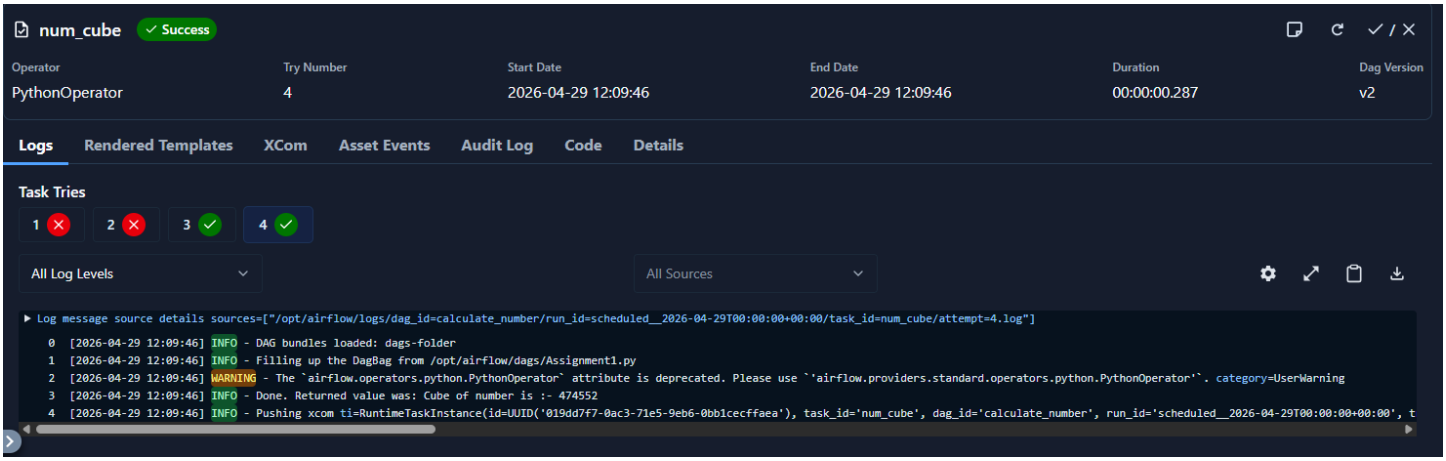
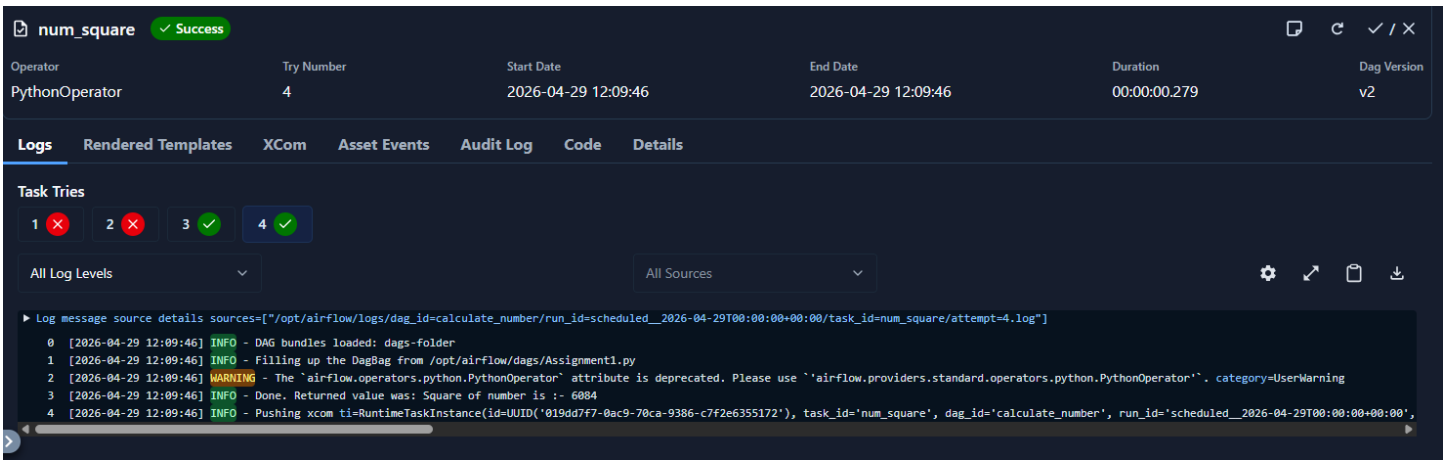
Results :-



Generated Number :-



Cube of Generated Number :-



Square of Generated Number :-